

The qualified instrument emerged through falsifiable gates
Failures, calibration, and replication are distinct—not an inevitable path

METHOD DECISION

One-shot L17 careful-minus-direct controller; plus, minus, orthogonal random

Separate source/readout from causal control; advance paired-mean prompt-boundary L27

Prospective safety/behavioral dose calibration; no accuracy or G/C/D selection

Hold controller fixed; isolate one-shot versus sustained duration

Replicate exact L27 controller at full dose on fresh items

Fresh independent evaluation of the fixed selected D75 controller

No detectable semantic movement
MICRO_Q1_NO
DETECTABLE_SIGNAL
N=50

Duration contrast observed; primary movement threshold not reached
GATE5_DURATION
EFFECT_BELOW
MOVEMENT_THRESHOLD
N=60

Large specific signal; historical safety classification remains destructive
GATE6_3_SINGLE_MEAN
DESTRUCTIVE
N=60

Movement replicated; relative validity/evaluability guards failed
GATE7_DESTRUCTIVE
N=120

Frozen lowest-eligible-lower-dose rule selected D75
GATE8_SAFE_LOWER
DOSE_SELECTED
N=50

Strong safe selected-dose development replication
GATE9_STRONG_SAFE
SELECTED_DOSE
REPLICATION
N=100

OBSERVED RESULT

Gate 4

Gate 5

Gates 6–6.3

Gate 7

Gate 8

Gate 9